

## Unit Testing

| ID | Purpose                                                                                                           | Steps                                                                                                                                                                 | Conditions                                                                                                 | Expected results                                                                | Outcomes |
|----|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------|
| P1 | Testing the Note/Signal class to ensure the frequency can be correctly stored                                     | Create a note or signal object with a valid frequency within the expected range of 1-1000 Hz and attempt to recall the value                                          | Let Frequency of note or signal equal 440 hZ                                                               | 440Hz should be returned                                                        | Pass     |
| E1 | Testing edge case of the note/signal class for storing frequency                                                  | create a note or signal object with a frequency of zero and attempt to recall the value                                                                               | frequency of note or signal must be equal to 0Hz                                                           | 0Hz should be returned                                                          | Pass     |
| E2 | Testing the note/signal class for when a error value may be inputted                                              | create a note or signal object with a negative frequency value and attempt to recall the value                                                                        | frequency of note or signal must be equal to -100Hz                                                        | frequency should be converted to a positive value and 100Hz should be returned  | Pass     |
| P2 | Testing the not/signal class with a very high frequency                                                           | Create a note or signal object with a very high frequency in case the filter fails and attempt to recall the value                                                    | frequency of note or signal must be equal to 20,000Hz                                                      | frequency should be stored correctly and the 20,000Hz should be returned        | Pass     |
| P3 | Testing the signal class to ensure the frequency can be updated                                                   | create a signal object with a valid frequency then attempt to use the SetFrequency method to update the frequency recalling the frequency both times                  | Initial frequency =100Hz then update it to 110Hz                                                           | 100Hz returned the first time and 110Hz the second time                         | Pass     |
| F1 | Testing for type errors within the input for the signal/note class                                                | create a signal or note object with a different data type than an integer and attempt to recall the frequency                                                         | input is "AJ2507"                                                                                          | program should return a type error                                              | Pass     |
| P4 | Testing the scale class with a correct input to ensure function is correct                                        | Using a set of correct note object with a list of corresponding notes as chars create a scale class and attmpt to recall the first and second note frequency and name | Use the frequencies , 82.41Hz, 110Hz,146,83Hz,196.00Hz,246.94Hz,329.63Hz and the note names , E, A,D,G,B,E | 82.41Hz,E and 110Hz , A should be returned from the program                     | Pass     |
| E3 | Testing edge case of the scale for when an empty scale is input                                                   | create a scale with 2 empty lists as inputs and attempt to recall a note                                                                                              | use empty lists as inputs                                                                                  | object should be created but nothing should be returned                         | Pass     |
| F1 | testing the scale class for an incorrect input                                                                    | create a scale with 2 lists of different lengths as inputs                                                                                                            | use the frequencies , 82.41Hz, 110Hz,146,83Hz,196.00Hz with the note names E,A,D,G,B,E                     | Object should not be created and return an aproprite error message              | fail     |
| P5 | Testing the find string method within the note analyser class with an expected input at an exact string match     | create a signal class with and scale class and run the find string method outputting the results                                                                      | use the signal frequency 82.41Hz                                                                           | should return the corresponding string to the inputted frequency with no errors | Pass     |
| P6 | testing the find string method within the note analyser class with an expected input that does not match a string | create a signal class and a scale class , then run find string method outputting the results                                                                          | Use the signal frequency 78.3                                                                              | should return the string label to the closest frequency within the scale        | Pass     |

Integration testing

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| E4     | test the find string method with a very low frequency input                                                                | create a signal object and a scale object then run the find string method outputting the results                                                      | Use the signal frequency of 60Hz                                                                                                                                                                                        | should return the string in the scale with the lowest frequency                                                                                                                         | Pass |
| E5     | test the find string method with a very high frequency input                                                               | create a signal object and a scale object then run the find string method outputting the results                                                      | use the signal frequency of 20,000Hz                                                                                                                                                                                    | should return the string with the highest frequency                                                                                                                                     | Pass |
| P7     | testing the find error method within the note analyser class for a signal that is in tune                                  | create a signal object and a scale object then run the find error function outputting the results                                                     | 82.41Hz                                                                                                                                                                                                                 | should return a 0 followed by "=" to indicate in tune                                                                                                                                   | Pass |
| P8     | testing the find error method within the note analyser class for a signal that is lower than the tolerance for the string  | create a signal object and a scale object then run the find error method outputting the results                                                       | Use a signal Frequency of 81.00Hz                                                                                                                                                                                       | should return the difference between the frequencies within the string followed by "b" indicating the note is flat and the signal is to low                                             | Pass |
| P9     | testing the find error method within the note analyser class for a signal that is higher than the tolerance of the string. | create a signal object and a scale object then run the find error method outputting the results                                                       | the frequency of the signal must be 83Hz                                                                                                                                                                                | should return the difference between the frequencies within the string followed by "#" indicating the note is sharp and the signal is to high                                           | Pass |
| E6     | testing the intune boundary of the program to see how close the signal frequency must be to output intune                  | create a signal object and a scale object then run the find error method outputting the results                                                       | the signal frequency must be 82.2Hz                                                                                                                                                                                     | should return the error followed by an = indicating the signal is in tune                                                                                                               | Pass |
| P10    | Check real time running function of the whole system                                                                       | run the program with a real time updating frequency signal that starts too flat then in tune then up too sharp and monitor the output as this changes | starting frequencies need to be 81Hz and the end frequency needs to be around 83Hz more than the target frequency. the changes need to be every 0.25s at steps of 0.25Hz as a suitable level of response for the system | expecting the error output to start at 2.41 with a "b" indicating flat tone then increase up to 0 with a "=" to indicate in tune finally finishing on 0.59 with a "#" to indicate sharp | Pass |
| AJ2507 |                                                                                                                            |                                                                                                                                                       |                                                                                                                                                                                                                         |                                                                                                                                                                                         |      |